1 A thin card is inserted between two separate iron cores. A coil is wound around one core as shown in Fig. 1.1.

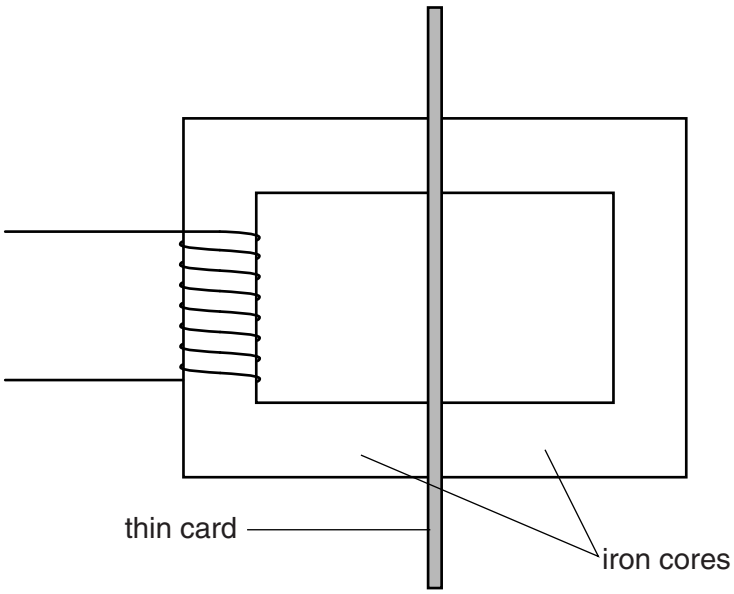

Fig. 1.1

A current in the coil may induce an e.m.f. in another coil wound on the other core. The induced e.m.f. V depends on the thickness t of the card.

A student suggests that

$$V = V_0 e^{-\sigma t}$$

where V_0 is the induced e.m.f. without card between the cores and σ is a constant.

Design a laboratory experiment to test the relationship between V and t and determine the value of σ. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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